Section 1: Introduction

The convergence of artificial intelligence (AI) and gene editing technologies, particularly with CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats/Associated Protein 9), marks a critical juncture in the evolution of biomedical innovation.[1] Discovered in 2012, CRISPR is a gene-editing technique that enables precise, targeted modifications to DNA, making it one of the most powerful tools in modern molecular biology [2]. While AI has drawn acclaim and skepticism, biotechnology's parallel revolution has received far less public recognition [3]. Although both fields have advanced rapidly, their integration now reshapes how genetic knowledge is produced, validated, and applied in practice, as AI increasingly aligns biological manipulation with computational prediction [4]. AI enhances CRISPR's technical precision, expanding its genetic engineering applications across agriculture, pest control, and disease prevention [5]-[7]. [2] As a result, the fusion of AI and CRISPR has prompted divisive discourse, encompassing therapeutic aspirations and commercial interests on one hand, and ethical concerns, political tensions, and moral unease on the other [3].

Optimists regard CRISPR as a "revolutionary breakthrough," with AI poised to "turbocharge" its therapeutic potential to cure genetic diseases, once thought to be impossible [7]-[9]. Supporters argue the combination of AI and CRISPR could transform healthcare systems and alleviate disease burdens, especially in historically neglected and under-resourced regions [2], [10], [11]. Yet this promise is accompanied by controversy, raising pressing ethical and political questions. Critics argue AI-driven genetic engineering is prohibitively expensive, dangerously unpredictable, and fraught with existential risk [12]. Beyond ethical and financial barriers, the development and governance of CRISPR, now amplified by AI, remains largely concentrated in the Global North, raising urgent concerns about health equity [14], [15]. [3] In this vein, AI-CRISPR is following a familiar path in which biomedical advances are concentrated in wealthy nations, while low- and middle-income countries (LMICs) remain excluded with unfulfilled promises of benefit and persistent barriers to access, reinforcing existing disparities in global health equity [14]-[16].[4]

For over a decade, CRISPR has been both celebrated and criticized for its potential implications for medicine and global health; however, AI's recent integration into the tool has

[1] Gene editing is a technique that allows scientists to precisely modify a short segment of DNA (gene). DNA contains genetic information to regulate living organisms [1].
[2] Genetic engineering refers to modifying organism's DNA by inserting or deleting genes [1].
[3] The Global North/West refers to wealthier, industrialized nations (e.g. U.S., Europe) while the Global South includes developing regions (e.g. Africa, South America) [13].
[4] For readability, this dissertation uses *AI-CRISPR* to refer to CRISPR gene editing enhanced by AI. Depending on context, CRISPR is described as a tool or biological system; *AI-CRISPR* here encompasses both.

specifically intensified concerns around data access, regulatory authority, and therapeutic inclusion [17]. While AI improves the accuracy and scalability of CRISPR applications, it also can inherit the biases present in training data and reflect structural asymmetries in research practices and infrastructure [18], [19]. Biotechnology and AI already embody global imbalances in access and use; their combination in AI-CRISPR risks compounding healthcare inequalities, depending on how the technology is developed, governed, and distributed [20].

This dissertation investigates the deployment of AI-CRISPR for Sickle cell disease (SCD) in sub-Saharan Africa (SSA). SCD is a severe hereditary hemoglobin disorder with high regional morbidity in SSA — up to 90% of affected children die before five — and it's the only genetic disease for which a functional CRISPR-based cure currently exists [21].[5,6] Despite being genetically well understood and regionally prevalent with 225,000 births a year, SCD has long been neglected in global health research and therapeutic solutions [24]. Given that gene editing remains a largely Western-led endeavor, now increasingly reliant on AI models trained on biased data, efforts to "bring" AI-CRISPR tools to the Global South, specifically to treat SCD in SSA, risk replicating a colonial logic in which scientific authority, technological design, and data ownership are centralized in the Global North [25]. Although SSA is internally diverse, this analysis adopts a regional framing to examine how transnational power structures shape health innovation — understood as new knowledge, tools, or practices shaped by social and institutional dynamics — across the region. In this model, African populations are framed not as equal partners, but as secondary targets of intervention [26].

Drawing on the concept of epistemic sovereignty — defined as the capacity of communities and nations to shape the production, validation, and governance of knowledge on their own terms — this dissertation argues that despite its technical promise, AI-CRISPR systems may perpetuate historical patterns of epistemic exclusion, extractivism, and Western dependency in SSA through algorithmic opacity and externally imposed governance frameworks. [7,8] In addition, this lens foregrounds questions of authority, legitimacy, and inclusion in scientific knowledge production, enabling an assessment not only about what AI-CRISPR can do, but also who defines its purpose and governs its use within the broader postcolonial landscape of biomedicine [29]. The core research questions guiding this inquiry are:

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[5] Hemoglobin refers to blood [22].

[6] Over 300,000 children are born with SCD worldwide, more than 75% of these births occur in SSA [23].

[7] Epistemic exclusion occurs when communities are denied credibility or recognition within dominant institutions [27], [28].

[8] Extractivism refers to exploitative extraction of resources, data, or knowledge that reinforces colonial hierarchies [25].

1. In what ways does the deployment of AI-CRISPR for SCD in SSA reproduce or contest historical asymmetries in data ownership, infrastructural access, and ethical authority?
2. How does the convergence of AI and CRISPR reshape global patterns of knowledge production, technological governance, and health equity?
3. Will AI-CRISPR reinforce or transform epistemic sovereignty in global gene editing, particularly in the context of SCD interventions in SSA?

To address these questions, this dissertation adopts a transdisciplinary and critical approach, informed by insights from postcolonial STS, global health equity concepts, and AI ethics principles to analyze AI-CRISPR technologies as sociotechnical systems shaped by existing power structures. Rather than treating CRISPR and AI as neutral and autonomous advances, this analysis situates these technologies within global histories of colonialism, exclusion, and scientific marginalization. Moreover, the case study of SCD in SSA provides a timely and relevant example to interrogate how frontier biotechnologies are governed, and for whom. Lastly, by centering epistemic sovereignty to highlight how power over knowledge production, algorithmic design, and genetic technologies is globally distributed, this dissertation contributes to a growing body of literature on AI and global development, particularly in the context of disease treatment. It offers a distinct contribution by focusing specifically on how AI's convergence with gene editing technologies impacts global health equity, an intersection that remains underexplored in current scholarship.

Overall, AI-CRISPR undoubtedly marks an important scientific milestone. Concurrently it represents a critical ethical inflection point: genomic innovations must be made genuinely accessible at every stage of development and implementation to the populations most affected by the diseases they aim to treat. Ultimately, this dissertation challenges techno-determinist narratives that frame such technologies as inherently transformative or neutral.[9] Instead, it argues that ethical governance of AI gene editing technology must account for the historical and structural conditions that shape global knowledge and power. Without reconfiguring the epistemic and institutional foundations of AI-CRISPR's design and deployment, this technology risks reinforcing the very exclusions it purports to solve.

The paper will proceed as follows. Section 2 outlines the scientific development of CRISPR, its integration with AI, key debates on how these technologies intersect with competing visions of knowledge production, infrastructure, and health equity. Section 3

[9] Techno-determinist refers to technology driving social change in a linear, value-neutral way, that obscures contextual factors [30].

introduces the theoretical framework, centering on epistemic sovereignty and its grounding in postcolonial STS and critiques of algorithmic governance. Next, Section 4 outlines the methodological approach which dovetails an interpretive analysis of scholarly literature, policy, and public discourse with a regional case study. Section 5 presents the empirical case of AI-CRISPR development and interventions in SSA for SCD. Section 6 offers a critical discussion of its implications for equity, legitimacy, and global health futures. Finally, Section 7 concludes by reflecting on the conditions under which AI-CRISPR technologies could contribute to, rather than compromise, more inclusive models of innovation in SSA — models grounded in epistemic sovereignty, local agency, and reconfigured power structures.

Section 2: Conceptual Foundations and Literature Review: CRISPR, AI, and Global Health Equity

This chapter maps the conceptual and contextual landscape of CRISPR and its evolving intersection with AI, situating both within broader trends in biotechnological innovation. It traces the evolution of genetic engineering, explores AI's integration into CRISPR, and presents key debates on its potential uses, societal implications, and impact on global development and equity. Drawing on relevant scholarship, this section critically examines how AI-CRISPR systems are entangled with structural asymmetries in knowledge production and access. Together, these insights underpin the dissertation's central inquiry into whether AI-CRISPR reproduces or contests global health inequities, particularly in the case of SCD in SSA.

2.1 Historical and Technical Background: The Evolution of CRISPR and Genomics

CRISPR is widely regarded as a landmark breakthrough in gene editing by geneticists, biotech investors, and medical researchers due to its accuracy, affordability, and programmability, often compared to "miraculous molecular scissors" [31]-[33]. However, its emergence rests on decades of molecular and genomic research [34]. The "first wave" of genetic engineering began in the 1950s with the work of Watson, Crick, and Franklin on the structure of DNA, enabling gene-level analysis of traits and diseases [35]. This was followed by recombinant DNA technology in the 1970s, which allowed interspecies gene transfer and prompted early bioethical debate, notably at the 1975 Asilomar Conference [36], [37]. In the late 1990s, scientists broadened DNA experimentation to include plants, bacteria, and animals [38].

The "second wave" of genomic innovation began with the Human Genome Project (HGP), which published the first complete human genome in 2001 [36]. The HGP pioneered advances in sequencing and mutation-tracking technologies, enhancing the ability to monitor genetic diseases [36]. As Reardon and Benjamin argue, the HGP not only transformed sequencing technology, but also redefined the political economy of genetics by turning DNA into commodified data assets for use in research and industry [39], [40]. These developments laid the scientific groundwork for genetic engineering technologies like CRISPR, a system derived from bacterial immune defenses and refined by Doudna and Charpentier for targeted gene editing [41]-[43].

Compared to older tools like zinc finger nucleases and TALENs, CRISPR is more modular, scalable, and precise, enabling direct DNA editing at targeted loci [19], [44], [45]. [10] Specifically, CRISPR's ability to efficiently "cut and paste" genetic material has made it the most widely adopted gene editing tool in biomedical research and synthetic biology, as it allows direct DNA sequence modifications [47], [48]. As Randhawa and Sengar note, CRISPR's effects now permeate almost every field of life sciences and has "revolutionized research" [19], [37], [44], [45]. Outside of the lab, CRISPR's therapeutic applications include somatic editing (e.g., for SCD), while germline interventions, altering embryos or reproductive cells, remain ethically contentious and legally restricted [49], [50].[11]

Nonetheless, CRISPR's broad applicability has coincided with a wider shift in the gene editing landscape, as biotechnologist Dr. Sidharthan emphasizes. The field has entered a "third wave" of genomic innovation, marked by accelerating clinical translation and commercial growth [52]-[54]. This growth is often cited in industry reports and policy papers as evidence of CRISPR's transformative impact, yet some scholars caution that market momentum may outpace ethical oversight [55], [56]. Still, the global market for gene editing — now worth over $1.72 billion — continues to expand, driven by breakthroughs in precision medicine, cell therapies, and advanced sequencing [57].[12] CRISPR has played a central role in this momentum: as gene editing becomes more cost-effective and scalable, CRISPR-enabled therapies are moving from experimental research into clinical practice [59]. Another branch of genetic engineering, gene therapy, illustrated in Fig. 1, has also grown in prominence with more than 3,900 therapeutic trials approved globally [45], [60].[13] Though scholars like Shawna Benston caution against inflated "hype" around CRISPR's growth, given its expanding applications and ongoing technical refinement, CRISPR remains a highly dynamic and consequential area of study, as geneticist Fyodor Urnov emphasizes [62], [63].

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[10] CRISPR uses guide molecule (RNA) to locate a specific DNA sequence before making a precise cut (using a Cas9 enzyme), allowing scientists to modify/repair the gene [46].

[11] Somatic edits target non-reproductive cells, meaning genetic changes only affect individual [50].

[12] Precision medicine as tailored healthcare based on your genes and lifestyle [58].

[13] CRISPR is a gene editing technique that's applied in gene therapy [61].

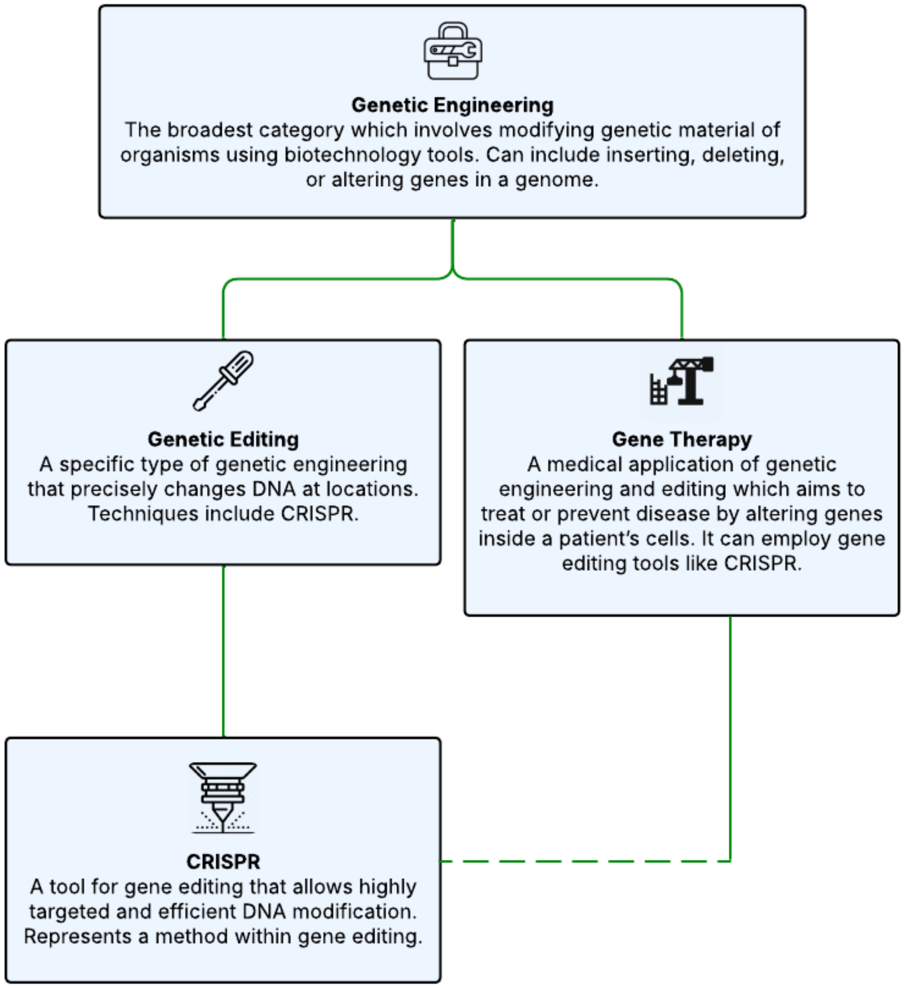

Fig. 1: Hierarchy of genetic engineering terminology [64].

2.1.1 Public Narratives, Ethical Imaginaries, and the Politics of Hype

As CRISPR nears routine clinical use, its potential has sparked diverse opinions and recommendations on its development, application, and regulation. For example, biotechnology professor Hank Greely argues CRISPR is best suited for non-human purposes, while others, including Harvard biologist David Liu and geneticist Kiran Musunuru, imagine CRISPR as a panacea to cure the full spectrum of genetic diseases [64]-[67], [69]-[73].[14] CRISPR co-creator Jennifer Doudna naturally emphasizes CRISPR is transformative for therapeutics, but calls for careful oversight in its expanding use [75]. Renowned geneticist George Church envisions more expansive CRISPR applications in all areas of life sciences, including human enhancement; conversely, Françoise Baylis critiques such techno-

[14] Non-human CRISPR applications include boosting crops, restoring ecosystems, and controlling pests [74].

optimism, arguing that dominant "science-first" narratives obscure critical ethical questions around accessibility, governance, and justice [76], [77].[15] Meanwhile, patients like Victoria Gray, one of the first to receive SCD CRISPR-based therapy in the US, embody both the promise and the uncertainty of gene editing, drawing attention to lived experience and access [79]. In this context, several scientists advocate for simplifying and scaling up CRISPR technologies to make them more accessible, while skeptics argue healthcare efforts should shift away from CRISPR and towards strengthening existing medical treatments, given the significant barriers to advanced biotechnology in the Global South [80]. These perspectives reflect a broader plurality of visions surrounding CRISPR's future. So, while some portray it as a vehicle for medical equity and healthcare democratization, others warn it could exacerbate social inequalities or open the door to eugenics and ecological disruption [81]-[86]. Dr. He Jiankui's 2018 announcement of the first gene-edited babies crystallized these tensions, prompting regulatory backlash after sensationalist media coverage [87].

While the He scandal catalyzed important discussions about gene editing, such events have polarized public discourse, reducing CRISPR to a binary of promise versus peril that risks obscuring the contextual, political, and epistemic dimensions [88]. As a result, attention to the ethical complexities of real-world clinical deployment, particularly in global contexts marked by uneven infrastructure, regulatory asymmetries, and limited access to care, is growing absent from these debates [88], [89]. While no clear consensus exists on CRISPR's future, and only one CRISPR therapy has been approved for clinical use in select countries, it's necessary to recognize the technology is quickly maturing, especially with AI embedded into its workflows [90]. Therefore, concerns around optimization, algorithmic design, and governance are increasingly important; the next section explores how AI is optimizing CRISPR's capabilities, introducing new technical, ethical, and epistemic challenges.

2.2 CRISPR and AI

Though CRISPR originated as a biological tool, its growing research and clinical applications increasingly rely on AI. Machine learning algorithms now support key stages of CRISPR workflows, including guide RNA selection, off-target risk prediction, and the identification of novel CRISPR-associated enzymes; Fig. 2 showcases several CRISPR studies implementing AI techniques [91], [92]. Neural networks, such as those used in DeepCRISPR and CRISPR-Net, model biological processes and improve design accuracy while reducing costs of wet-

[15] Belief that emerging technologies will drive progress and are key to solve complex societal problems [78].

lab experimentation [93], [94]. As such, these AI techniques help streamline technical tasks and improve variant interpretation and gene detection [95], [96]. Essentially, AI enhances CRISPR's precision by enabling targeting specific genetic components, while minimizing unintended effects [8], [93]. Recently, Doudna emphasized AI's capacity to rapidly process vast genomic datasets will propel gene-editing research and ultimately allow for safer clinical trials [9]. Scholars like Milad argue AI is not merely theoretically or discursively hyped, but central to how CRISPR research advances today [97]. Overall, AI complements CRISPR's current capabilities, and will supplement its future applications by providing precise and personalized analytics that expand research and reach, illustrated in Fig. 3 [94].

Author	Contributions
Li et al. (2021)	Proposed an end-to-end deep learning framework named CROTON, an approach used deep multi-task convolutional neural networks and neural architecture search (NAS) to automate both feature and model engineering
Liu et al. (2022)	Developed Apindel, a deep learning model employing BiLSTM and Attention mechanism, to predict a comprehensive range of Cas9-generated mutational outcomes, surpassing previous models in accuracy
Wang et al. (2021)	Introduced EditPredict, a CNN-based model that accurately predicts RNA editing events in humans, including those resulting from CRISPR-Cas9 knockout of the ADAR1 enzyme
Li et al. (2022b)	Devised SeqGAN, a model combining CNN and an adversarial network, to predict CRISPR off-target cleavage sites, achieving superior performance in cross-validation tests
Li et al. (2022c)	Proposed machine learning approaches, including deep learning, for identifying human blood cells with CRISPR-mediated fetal chromatin domain (FCD) ablations, with promising results in the prediction of edited cells
Naert et al. (2020)	Applied predictive modeling of editing outcomes to maximize CRISPR-Cas9 phenotype penetrance in *Xenopus* and zebrafish embryos, improving phenotype penetrance in the F0 generation
Naert et al. (2021)	Introduced CRISPR-SID, a method using deep learning to predict double-strand break repair patterns for identifying genes essential for tumorigenesis in a *Xenopus* tropicalis desmoid tumor model
Marquart et al. (2021)	Developed BE-DICT, a deep learning model incorporating attention mechanisms, to predict base editing outcomes, providing a versatile tool for genome editing
Zhang et al. (2021b)	Conducted a comprehensive evaluation of the editing efficiency of several Cas9 variants and utilized a deep learning model to verify and predict SpRY off-target sites, informing the refinement of Cas9 variants for precise editing

Fig. 2: Lee's review of AI deep learning techniques in CRISPR-Cas9 research [91].

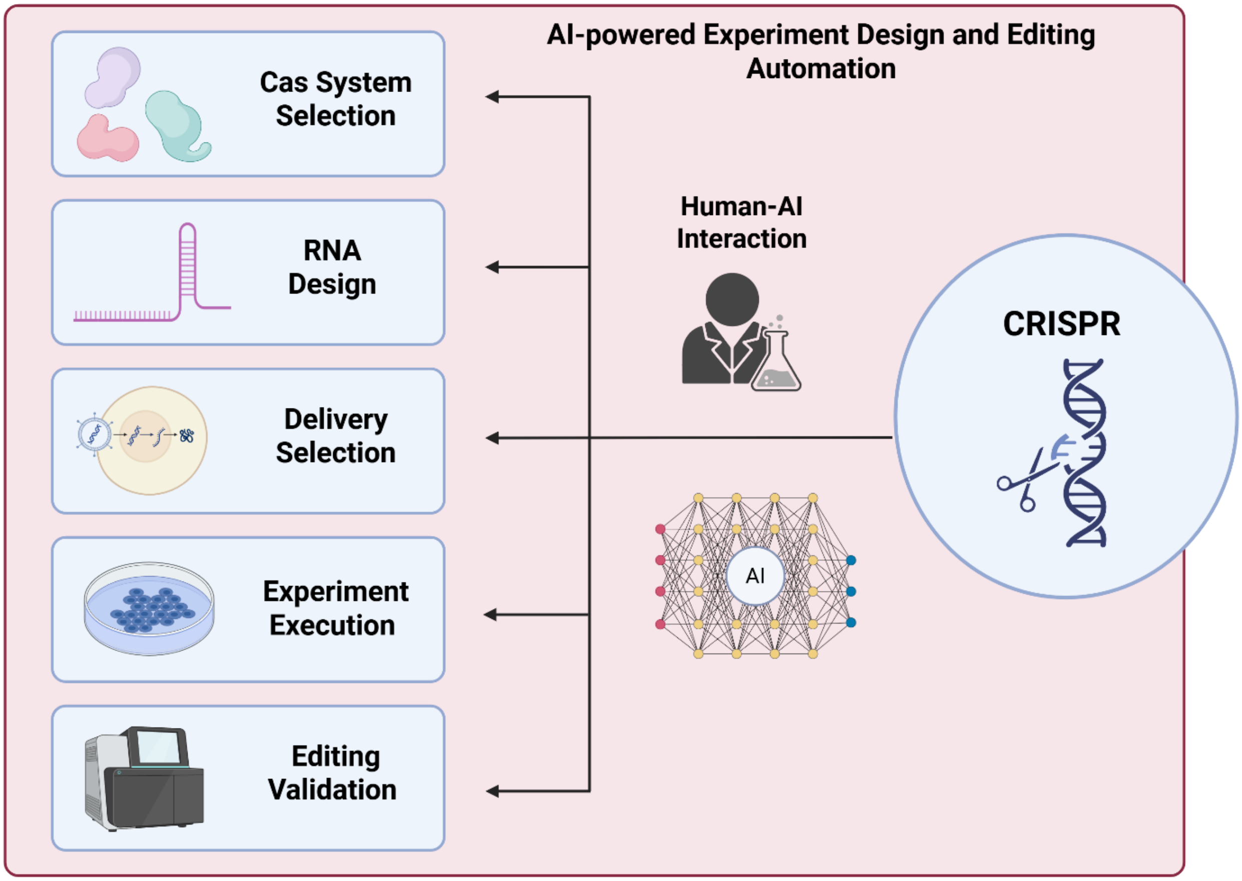

Fig. 3: AI in CRISPR [17].

Although many scholars place hope in AI-CRISPR's technical sophistication to make gene editing cheaper, safer, and thus more accessible, incorporating AI into CRISPR further complicates the existing governance and representation challenges of the technology [8], [9], [98]. AI augments CRISPR's capabilities but also shapes how CRISPR systems interpret data from diverse populations, predict therapeutic outcomes, and make targeted gene edits [99]. Consequently, this computational shift creates specific complications for gene editing interventions and governance in LMICs, as AI systems developed in the Global North may fail to account for local genomic diversity, health priorities, or regulatory contexts [100]. These dynamics underscore the need to situate AI-CRISPR within the broader global asymmetries that shape gene editing innovation, a topic explored in the next subsection.

2.3 Global Asymmetries of Gene Editing and AI-CRISPR Innovation

As studies from Dixit et al. and Doxzen et al. stress, AI will influence the future of CRISPR and gene editing, but it will be differences in global research capacity, funding, and ethical oversight that ultimately determine how the technology is applied and who benefits [94], [101]. As Masood, Dunn, and Pang convey, the global conversation on genome editing is anything *but* global [102]-[106]. Despite promises of "globally transformative potential," the benefits of AI-CRISPR systems remain heavily concentrated in a narrow band of high-income countries [74], [84]. This concentration reflects broader structural dynamics, as contemporary norms for gene editing are largely defined by institutions in the Global North, which dominate research, funding, technological pipelines, and regulatory architectures [90], [91]. For instance, as of 2023, over 70% of CRISPR clinical trials were based in five high-income countries, led by U.S. and European biotech firms [108], [109].

The Global North effectively controls every stage of development in the AI-CRISPR pipeline, from agenda-setting, where research priorities are shaped by Western funding institutions, to scientific inquiry, when AI models and gene-editing frameworks are calibrated according to datasets disproportionately sourced from Eurocentric populations [110]. This process is then codified by Northern regulatory regimes advised by biotech firms and reinforced by intellectual property systems that limit access and commercialization, keeping biotechnological ownership and decision-making largely tethered to the Global North [111], [112].[16]

Such dominance is not merely logistical or technological, but epistemic as well: it defines which knowledge is relevant, what risks are considered, and whose health needs are prioritized [5], [28], [101]. Focusing on epistemic sovereignty is essential in the context of AI-gene editing, as it highlights how authority over scientific agendas, data governance, and normative frameworks remains unevenly distributed — shaping not only who generates genomic knowledge, but whose futures are constructed and legitimized through its application. As Jasanoff and Hurlbut emphasize, "*we must ask not only what can be done, but what should be done, by whom, and with whose consent,*" challenging techno-determinist assumptions and generalizations that position gene editing as ubiquitously beneficial [104]. Instead, these scholars call us to scrutinize the imaginaries guiding AI-CRISPR, and whose futures it serves [113].[17]

By adopting more a sociotechnical view of gene editing, this dissertation contends the effects of AI-CRISPR will be contingent on institutional contexts, interpretive narratives that surround its use, and the uneven distribution of risks and benefits across different

[16] Regime refers to system of rules, institutions, and practices [112].

[17] Imaginaries here refers to the collective vision of a desirable future shaped by science/technology [104].

publics [114]. These structural and epistemic asymmetries are not abstract; they have direct consequences for global health equity, especially in the Global South where populations face persistent healthcare disparities.[18] While Rasheed highlights that power dynamics in the Global South are historically layered, and albeit complex, the existing global health system, built on historical power and privilege, continues to restrain SSA LMICs from fundamentally shaping their own biomedical interventions [115]. Recognizing the epistemic dimension of this inequality is central to evaluating the global implications of AI-CRISPR. These issues are examined in greater detail in Section 5's case study on SCD in SSA.

2.4 Global Health Equity and the Promise of CRISPR

As discussed, many advocates of AI-CRISPR emphasize its potential to transform medicine, often assuming that its technical sophistication will naturally deliver better outcomes and broader public benefit [116]-[118]. Yet, as Parthasarathy notes, technological innovation does not automatically serve everyone [119]. In this vein, a substantial corpus of literature explores the impact of genomic technologies on health in the realms of education, law, and bioethics, but across thousands of CRISPR-related studies, relatively few examine its implications for health equity for underserved communities [19].

This gap requires attention, as AI-CRISPR's "innovative" promise is inseparable from considerations of global health equity: the fair, just opportunity for all populations to achieve the highest attainable standard of health [120]. Jooma remarks, defining health equity in genomics is challenging because clinical evidence is still emerging and translation into medical practice is relatively new [121]. As such, in this dissertation, health equity constitutes globally relevant genomic knowledge, fair access to services, and unbiased implementation, grounded in an understanding of how biological, institutional, and socioeconomic factors shape disease [121], [122]. Thus, advancing health equity through AI-CRISPR requires equal, effective, affordable access, with treatments free from biases like discrimination and underrepresentation — biases that reflect and reinforce systematic global health inequities.

With this understanding, this dissertation analyzes how AI-CRISPR risks reproducing patterns of epistemic exclusion that undermine health equity, keeping historically marginalized communities most affected by genetic diseases at the margins of research, decision-making, and benefit-sharing [123], [124]. This is particularly visible in the case of SCD: despite a high epidemiological burden in SSA, CRISPR therapies have been developed, trialed, and distributed almost exclusively in high-income countries, often using AI models

[18] Disparities include lower life expectancy, higher disease burden, and limited healthcare access [115].

trained on genomic data that underrepresent African populations [109], [125]. These omissions point to a broader failure of inclusion and highlight the need for frameworks that foreground local agency in knowledge governance. Epistemic sovereignty offers such a lens, calling attention to the need for AI-CRISPR systems that are aligned with ethical technical standards but also attuned to serve diverse local values, priorities, and forms of expertise [126]. The following section introduces epistemic sovereignty in greater depth, establishing it as the core theoretical framework through which the dissertation evaluates CRISPR-AI interventions in global health.

Section 3: Theoretical Framework

This chapter outlines the theoretical framework guiding the analysis of CRISPR and AI in global health innovation. This framework is anchored in the concept of epistemic sovereignty — the capacity of communities and nations to govern the production, interpretation, and application of knowledge on their own terms [127], [128]. In the context of AI-gene editing, epistemic sovereignty provides a critical lens to examine how authority over data, scientific standards, and algorithmic models is unevenly distributed across global systems. This analysis is supported by traditions in postcolonial STS, as well as critiques of algorithmic governance, which help historicize and contextualize how colonial epistemic hierarchies shape scientific authority, and how these embedded dynamics impact computational genomics today [129].[19] Together, these theoretical strands help interrogate how AI-CRISPR technology reflects and reinforces power by privileging dominant actors who control the production and governance of scientific knowledge.

3.1 Epistemic Sovereignty

Originally a concept in political philosophy and international relations, sovereignty traditionally refers to control over territory, law, and political authority [130]-[132]. Epistemic sovereignty extends and reinterprets this logic to the domain of knowledge production: it concerns the power to shape what counts as valid knowledge, who gets to produce it, and how it is interpreted and applied [29], [131], [133]. In the realm of STS, epistemic sovereignty demands more than inclusion within dominant paradigms; it calls for structural authority over scientific agendas, methods, and interpretive frameworks [29], [134]. Scholarly debates accentuate the fluidity of this concept across disciplines, but largely emphasize the longstanding centrality of political dynamics in shaping knowledge diffusion within science [29], [135].[20]

Though a relatively nascent idea in the context of biotechnologies specifically, epistemic sovereignty offers an effective grounding to examine how scientific authority and governance around gene editing remain unequally distributed as the field quickly advances. Here epistemic sovereignty functions both as an analytic concept that reveals how power shapes knowledge production, and as an evaluative framework that suggests how systems should be restructured [137]. It exposes current power imbalances and also advocates for

[19] Computational genomics applies AI analysis to uncover biological interactions from genomic sequences [129].

[20] Scholars have contrasting perspectives on how individuals acquire knowledge: Zagzebski endorses external epistemic authority via self-trust while Melnyk insists belief formation must be autonomous [136].

disrupting systems that marginalize communities [138]. Therefore, this lens serves to challenge the marginal role that LMICs occupy in global scientific systems, drawing attention to the structural exclusions embedded in current models of technological development: agendas, funding, and epistemic priorities are dictated by the Global North [123], [134], [139]. Scholars such as Prescod-Weinstein and Visvanathan have shown that despite the promotion of "axioms centered on equal global collaboration" by Western institutions, research communities and stakeholders in the Global South often remain epistemically subordinate — frequently tasked with data collection or clinical implementation but excluded from decision-making [140]-[143]. In genomic science specifically, this pattern manifests as underrepresentation in training datasets, insufficient regulatory oversight of CRISPR workflows, and the marginalization of localized ethical frameworks [144].

Epistemic sovereignty thus becomes essential for understanding how African institutions, researchers, and publics have historically been positioned as subjects or data sources, rather than as knowledge producers [145]. AI exacerbates these dynamics by encoding assumptions into models that reflect the values, priorities, and biases of their designers [18]. When used in gene editing, such systems risk not only technical inaccuracies but epistemic harm: they distort scientific relevance and undermine local agency [39], [146]. In this light, epistemic sovereignty serves both as an analytical lens and as an asset that SSA can embrace, challenging extractivist models by advocating for inclusive knowledge practices, regional priority-setting, and representative ethical authority around AI-CRISPR [146]. This perspective not only enables a comprehensive examination of its innovative lifecycle — from defining disease priorities and selecting data, to governing deployment and algorithmic interpretation — but also underscores SSA's potential to reclaim control over its scientific agenda to treat SCD, which will be discussed in more detail in Section 6.

3.2 Postcolonial Science and Technology Studies (STS)

To contextualize epistemic sovereignty, this dissertation draws on postcolonial STS, a field that examines how global scientific authority has been historically constructed through colonial extraction and epistemic erasure. While mainstream STS explores the social construction of scientific facts, postcolonial STS asks who is authorized to construct those facts, and how colonial legacies shape the modern global hierarchy of knowledge [147], [148]. As epistemic sovereignty centers on the right to govern knowledge production and interpretation, postcolonial STS interrogates the historical and structural conditions through which global hierarchies of knowledge have been produced, naturalized, and contested [149]. Postcolonial STS provides the analytic vocabulary and historical foundation to evaluate knowledge dominance, thereby bridging theoretical scholarship with critiques of emerging biotechnologies such as AI-CRISPR [124], [150]-[152].

For example, postcolonial STS elucidate that the “so-called universal, objective, and neutral practices” of contemporary healthcare are in fact, deeply influenced by the power structures that produced them [149]. Scholars such as Harding, Santos, and Mavhunga argue, scientific knowledge often delegitimizes Indigenous and non-Western epistemologies, centralizing Euro-American authority [153]-[155]. Similarly, Lumun documents how indigenous African healing systems were displaced by colonial medicine, eroding epistemic sovereignty and reinforcing dependency on the West [156]. These asymmetries persist today: regulatory models, treatment standards, and ethical frameworks in global health remain anchored in Northern institutions, regardless of geographic or cultural relevance [152].

Notably, these historical patterns of science and power do not exist in a vacuum; Harding challenges the aphorism of “objective science,” arguing that knowledge production is intrinsically linked to its social and political contexts [153]. For example, Anderson traces how colonial medicine portrayed tropical diseases as exotic threats to imperial actors, leading to treatments developed in the metropole and imposed on colonies to maintain order, rather than enhance Indigenous wellbeing [147], [157]. Pearson-Patel and Hayden similarly note that while the 2014-2016 Ebola outbreak primarily affected Africa, the international response prioritized Western care models and narratives steeped in colonial-era frameworks [158]-[160]. These histories exemplify that health sciences are neither neutral nor apolitical: medical interventions and disease responses have been shaped by the contexts in which they are created and deployed, an idea Cislaghi et al.’s research promotes [161]. This dynamic persists in genomics, with Northern institutions dictating methods, ethics, and now AI systems that govern genomic knowledge and legitimacy — a focus of the next subsection on algorithmic governance

Overall, postcolonial STS complements the central lens of epistemic sovereignty by historicizing the colonial roots of global scientific hierarchies, highlighting that the global domination of Western science is the result of historical power imbalances, not methodological superiority [127], [151], [155], [163]. However, while postcolonial STS offers utility in untangling the historical foundations of epistemic exclusion, it is the concept of epistemic sovereignty that veritably offers a forward-looking framework, affirming the right of communities to define and govern knowledge on their own terms [138]. In doing so, this approach avows the legitimacy of subaltern epistemologies and contests universalist scientific norms; Section 6 revisits this framing to examine how SSA communities can reclaim control over scientific agendas, ethical standards, and AI-CRISPR governance [164].

3.3 Algorithmic Governance: Global Health and AI

As AI takes on a greater role in CRISPR systems, it's reshaping the foundations of biomedical choices. While AI-CRISPR promises greater efficiency and speed, it also introduces new complexity: decisions once made by human expertise are now partially delegated to opaque, proprietary algorithms [166]. This shift elevates AI from an auxiliary element within CRISPR pipelines to a core component of gene editing, determining what forms of genomic knowledge are legible, what procedural risks are acceptable, and which populations are rendered visible within biomedical systems [167], [168]. These developments make algorithmic governance essential to understanding how AI redistributes epistemic authority within CRISPR systems. [21]

Mainstream AI governance and ethics frameworks typically emphasize principles such as fairness, transparency, and accountability [171]. While valuable, these approaches often focus narrowly on algorithmic performance and individual harms, overlooking the deeper structural and epistemic inequalities embedded in AI systems. In global health, for example, studies show algorithmic models are frequently trained on datasets from high-income settings, limiting their generalizability to populations in the Global South [111], [172], [173]. Moreover, uneven access to the tools, infrastructure, and evaluation metrics needed to develop or regulate AI raises urgent concerns about computational dependency [111], [163]. Existing governance models rarely address these asymmetries, nor do they meaningfully involve affected communities in defining design choices, evaluation standards, or accountability mechanisms, further marginalizing Global South perspectives in shaping genomic futures [163].

The patented nature of many AI-CRISPR systems compounds these challenges, limiting external scrutiny and complicating efforts to validate performance, assess risks, or demand accountability [172]. Furthermore, dominant AI governance frameworks are largely shaped by institutions and states in the Global North, consolidating control over the norms, benchmarks, and ethical priorities that define what constitutes "responsible AI" in genomic medicine [163]. As such, effective algorithmic governance must move beyond technical safeguards to confront the structural inequities that underlie AI-CRISPR development. This calls for models that embed epistemic justice, democratized oversight, and more inclusive participation in setting the standards that govern AI's role in gene editing.

AI-CRISPR systems are not only technically complex but also epistemically charged, embedding assumptions about biological normalcy, therapeutic viability, and acceptable

[21] Algorithmic governance is the study of socio-technical systems in which the governance function (social) is executed in the presence of an algorithmic model (technical) [169], [170].

risk for genomic therapeutic interventions [174]. Yet the decisions these systems enable, which have direct clinical consequences, are rarely co-produced with communities most affected by disease, SCD in this case [94]. The algorithmic governance of AI-CRISPR SCD therapies thereby raises urgent questions about technological dependency, validation, and legitimacy, demanding critical attention to how AI mediates genomic knowledge and authority in SSA.

3.4 Towards a Critical Framework

The confluence of CRISPR and AI marks a new biotechnological frontier, entangled in complex sociotechnical and political histories. To analyze this convergence, this dissertation adopts a critical framework grounded in epistemic sovereignty, informed by insights from postcolonial STS and notions of ethical algorithmic governance. Though each of these literatures offers distinct contributions, together they focus attention on how knowledge, legitimacy, and technological authority are constructed and contested, which is essential to analyze AI-CRISPR as a tool to either reinforce or recast health equity opportunities for SSA.

Epistemic sovereignty anchors the analysis: it centers questions of who produces, controls, and legitimates knowledge in genomic medicine. It enables an evaluation of how SSA publics and institutions are positioned in global biotechnology, not simply as beneficiaries, but as potential co-governors of AI-CRISPR systems. Postcolonial STS contextualizes this framework by tracing the colonial roots of exclusion. Algorithmic governance illustrates how AI systems mediate these exclusions in new ways, obscuring human accountability and reinforcing inequity through automation and opacity.

It's pertinent to mention that global bioethics, which examines health technologies across cultures, might seem like a useful point of entry; however, the concept is insufficient to study AI-CRISPR in the scope of this work.[22] Critics argue that global bioethics remains too abstract and universalist, aiming to apply ethical principles across diverse cultural, political, and social contexts [178]. This broad approach can obscure the lived realities of communities in the Global South, overlook historical injustices, and sideline regional epistemologies crucial for ethical gene-editing governance [178]-[180]. In doing so, it reinforces technocratic and neoliberal norms by perpetuating colonial knowledge systems, as scholars Richardson, Santos, and Selin all note [181]-[183]. In contrast, epistemic

[22] Beauchamp and Childress' seminal *Principles of Biomedical Ethics* outlines the core bioethics principles: autonomy, non-maleficence, beneficence, and justice [177].

sovereignty interrogates who controls and legitimizes genomic science, offering a stronger framework for analyzing AI-CRISPR's use, especially when informed by research in postcolonial STS and algorithmic governance.

This approach clarifies how deep-rooted exclusions in science and authority shape the global deployment of AI-CRISPR in LMICs in SSA, where patterns of biomedical exclusion, algorithmic bias, and regulatory dependency persist. Expanding access alone does not ensure equitable participation in AI-driven gene editing; it requires critical attention to how these biotechnologies are created, governed, and contested. While frameworks such as global bioethics offer valuable insights on inclusion and accessibility, this dissertation adopts epistemic sovereignty because it addresses the structural roots of epistemic inequality — an essential focus given AI's accelerating role in genomics. The next section outlines the methodology, followed by a case study that applies this framework to AI-CRISPR interventions for SCD in SSA

Section 4: Methodology

This section outlines the methodological approach used to examine AI-CRISPR interventions for SCD in SSA as a case study. The study adopts a qualitative, critical case study methodology, suited to exploring complex, real-world phenomena where context and subject are entangled, as Yin and Salmons highlight [184]-[186]. The method is interpretive and interdisciplinary by drawing on epistemic sovereignty literature, insights from postcolonial STS, and perspectives on fair algorithmic governance. In doing so, this approach helps cultivate a more holistic and nuanced analysis by integrating diverse insights across different fields [180]. Informed, in part, by elements of Glaser and Strauss' grounded theory, this analysis proceeds inductively, iteratively building conceptual insights from patterns across multiple sources rather than testing any predefined hypothesis about AI-CRISPR [187], [188].

4.1 Case Study Rationale: AI-CRISPR for SCD in SSA

SCD is a paradigmatic example of epidemiological neglect and biomedical promise. While its genetic etiology is well understood, SCD has long been marginalized in global health research, policy, and pharmaceutical investment, according to the WHO [189], [190].[23] This neglect is especially stark given a majority of SCD cases occur in SSA, where mortality and morbidity remain disproportionately high, making the region a particularly important to address. For instance, recent data estimated the region's share to make up over 79% of SCD cases worldwide [191]. Furthermore, SCD is the only condition with an approved CRISPR therapy, making it an ideal case for studying gene-editing interventions; for example, SCD CRISPR therapies like Casgevy™ (exa-cel) demonstrate a 96% success rate and have been approved in the US and UK [90]. So, while hailed as potential cures for SCD, these somatic gene therapies are developed, validated, and commercialized in high-income countries, despite SSA carrying the largest share of the global disease burden. This disparity underscores the urgent need to examine current patterns of science and authority to ensure that breakthroughs reach those who need them most.

This case was selected because it provides a fruitful site to investigate the dissertation's central concern: how advanced biotechnologies like AI-CRISPR are embedded within broader histories of colonial science and contemporary biomedical regimes, reinforcing epistemic exclusion and asymmetries in health equity [192]. The convergence of high disease burden, regulatory asymmetry, and emerging algorithmic

[23] SCD is caused by a "relatively simple" genetic mutation, making it easier to track and study.

infrastructures makes the SCD-AI-CRISPR triad in SSA a critical site for interrogating global inequities in gene editing technologies.

Although SSA is used as a regional frame, this analysis acknowledges the internal diversity across national African sociopolitical landscapes, institutional structures, and epistemic traditions. In this regard, this study does not claim representativeness across all settings, but rather traces how epistemic and institutional power are organized within and across transnational health and innovation systems.[24] Section 5's case study maps larger market trends, research patterns, and funding norms in AI-gene editing; SCD in SSA serves as a focal point for examining epistemic exclusion and questions of equity in biomedical innovation in the region.

4.2 Sources and Analytical Techniques

The analysis draws on diverse primary and secondary sources including peer-reviewed literature in bioethics, AI ethics, STS, and genomic medicine; policy documents from the WHO, NIH, and African Medicines Agency, and CRISPR firms; grey literature on SCD trials and regulatory debates; public statements by African scientists, bioethicists, and policymakers. Although no primary fieldwork was conducted due to time and logistical constraints, the analysis achieves empirical depth through data triangulation, combining multiple types of documentary and textual sources and focusing on how key actors frame problems, justify interventions, and establish ethical or technical standards [193]. Particular attention is given to the language of innovation, metrics of success, and patterns of inclusion or exclusion. This approach supports a comprehensive examination of how knowledge, power, and authority circulate in AI–CRISPR innovation.

4.3 Analytical Structure

Rather than separating theory and analysis, the dissertation adopts an embedded analytical structure, integrating the lens of epistemic sovereignty throughout the case study. Section 5 is organized thematically, with each subsection addressing a key phase in the AI–CRISPR deployment pipeline such as data curation, algorithm design, clinical translation, and regulatory decision-making. Each phase is used to examine how knowledge is produced, authorized, and governed, particularly in relation to African publics, institutions, and health

[24] Several country-specific analyses of AI-CRISPR were not viable given the scope of this dissertation's research. SCD is also recognized a larger problem of the SSA and often measured in regional statistics.

systems. This integrated structure reflects the entangled nature of epistemic, ethical, and infrastructural dynamics in contemporary biomedical innovation.

Section 5: Case Study: AI-CRISPR in SSA

This section contributes to the emerging analysis of AI-CRISPR systems by showing how AI operates as a critical subsystem that reshapes the development, deployment, and governance of gene-editing technologies. Using SCD interventions in SSA as a case study, it explores how AI-CRISPR systems both reproduce and reconfigure global asymmetries in knowledge production, infrastructure, and therapeutic authority. SCD, as the first human disease to receive licensed CRISPR-based therapy, provides a unique vantage point for analyzing these dynamics. Unlike much existing scholarship, which focuses on AI's technical contributions or CRISPR's bioethical concerns in high-income contexts, this section foregrounds the political, infrastructural, and epistemic dependencies that shape AI-CRISPR governance in SSA.

5.1 SCD in SSA: A Genetic Burden Amid Structural Neglect

SCD is the world's most common monogenic disorder, caused by a single point mutation in the β-globin gene that impairs blood cells oxygen-carrying capacity [194]. Over time, SCD leads to chronic pain, organ damage, and premature death [195]. While SCD affects populations worldwide, its prevalence is most acutely concentrated in SSA [21]. Exact diagnostic metrics vary due to limited reporting, the UN estimates that between 240,000 and 300,000 children are born with SCD in SSA annually, with a total prevalence of 12–15 million individuals [194]. In the absence of newborn screening and disease-modifying therapies, over half of affected infants die before age five, whereas in high-income countries, the median survival exceeds 40 years [196], [197]. Beyond its biomedical toll, SCD erodes national productivity and household economic security, reinforcing cycles of poverty [198].

Yet these health outcomes cannot be understood in quantitative biomedical terms alone. The postcolonial STS insights developed in this dissertation show that SCD's neglect in SSA is not incidental or purely technical. Rather, it is the product of long-standing patterns of unequal development and governance — where global health priorities, resource allocation, and research agendas have systematically marginalized SSA's needs [18], [25]. This context shapes how AI-CRISPR interventions for SCD are developed, deployed, and governed, raising critical questions about who controls genomic futures and on what terms.

SCD holds symbolic and practical significance for AI-CRISPR governance as the first condition targeted by a licensed CRISPR-based therapy [90].[25] While SCD treatment in SSA has largely focused on palliative care, AI-CRISPR offers a promising path to address the root

[25] Casgevy (exa-cel) was approved for human use in December 2023 [199].

cause of the disease as shown in Fig. 4, with the potential to help cure millions of patients in the future [200], [201]. Thus far, the populations most affected by SCD, particularly those in SSA, remain largely peripheral to these processes. SCD is therefore not merely a clinical challenge, but a politically charged case of epistemic and therapeutic neglect. SSA patients are doubly excluded from the biomedical innovations that could transform care, and from the systems that mediate their development and deployment. This raises critical questions about whether AI-CRISPR will reinforce existing global health inequities or be reconfigured to serve local priorities.

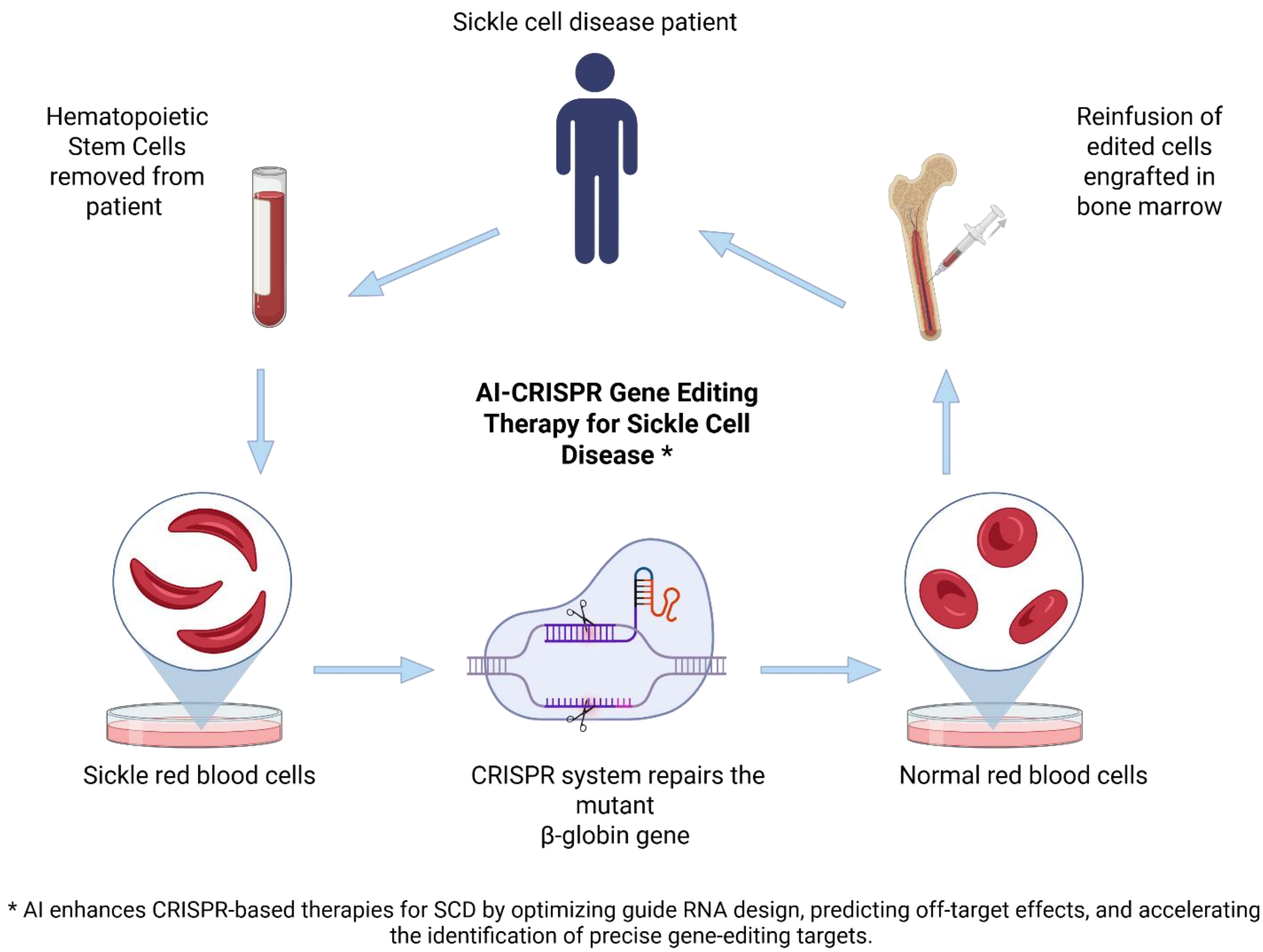

Fig. 4: CRISPR intervention for SCD.

5.2 The AI-CRISPR Pipeline: Global Innovation, Local Exclusion

Before the full CRISPR innovation pipeline even takes shape, from preclinical research to regulatory approval and market deployment, systemic barriers already limit the role that SSA institutions and researchers can play. Although a small number of African laboratories possess the capacity to engage in AI-CRISPR research, their ability to generate widespread impact for SCD patients in SSA is severely constrained by Western monopolies over intellectual property, manufacturing, and funding networks [202]. The following subsections trace how these asymmetries are embedded at each stage of the AI-CRISPR pipeline.[26]

5.2.1 Preclinical research: SSA's Peripheral Role in Knowledge Production

The preclinical development of AI-CRISPR therapies for SCD has been dominated by institutions in the Global North. Leading universities, including Harvard, Stanford, the University of Paris, and firms such as Vertex Pharmaceuticals and CRISPR Therapeutics control the technical design of CRISPR strategies and the AI models that optimize guide RNA selection and off-target predictions [204], [205]. As Rock illustrates, African researchers are largely positioned as collaborators providing biospecimens or data rather than as intellectual co-leads in CRISPR SCD [206]. In this vein, published analyses highlight a striking absence of African leadership in CRISPR sickle-cell research, with African institutions rarely appearing in senior authorship positions [207]. While specific data on CRISPR-SCD researchers in SSA is limited, Fig. 5 suggests that the underrepresentation of African authors in CRISPR research likely extends to SCD researchers. [27]

This exclusion is not just material but epistemic: positioning African scholars and institutions on the periphery limits regional input into research priorities, model assumptions, and ethical considerations, reflecting a structural exclusion that constrains SSA's epistemic sovereignty at the earliest stages of innovation. Notably, this silencing is not always explicit; it can be embedded in authorship practices, project framing, and peer review standards that reward alignment with existing paradigms of preclinical research [145], [206].

[26] The "pipeline" stages covered in this section represent the standard phases of a therapy or drug development pipeline [203].

[27] SCD is the only disease with active CRISPR treatments at this time, so metrics on SSA authorship in CRISPR research inherently include SCD studies.

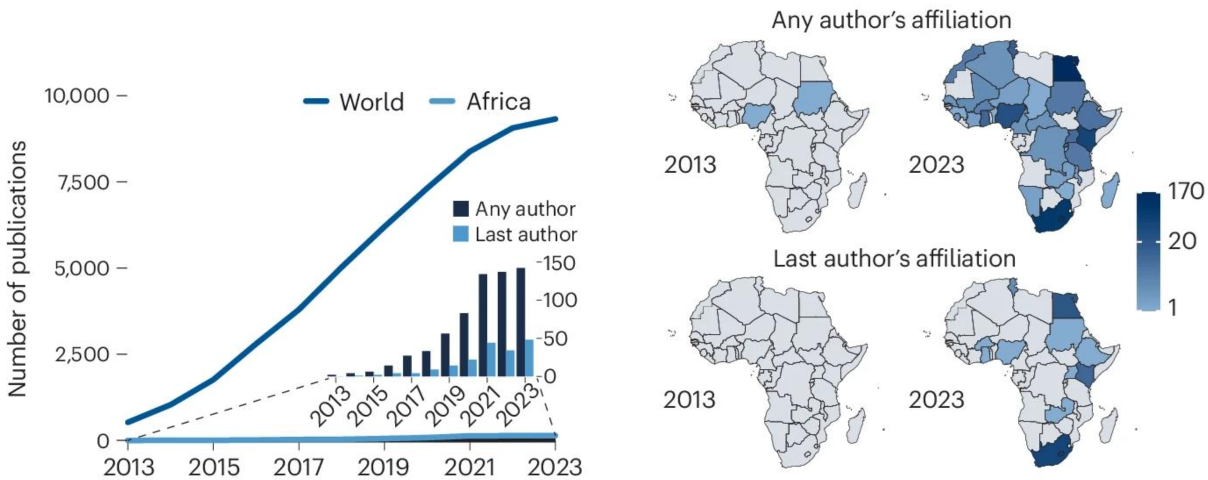

Fig. 5: Spread of authorship on CRISPR/gene editing [145].

5.2.2. Clinical Trials: The Absent African Patient

Next, clinical trials for CRISPR-SCD therapies, which largely shape the final market CRISPR product, have been conducted exclusively in the US and Europe. Instead, firms domiciled in the Global North including Vertex Pharmaceuticals, CRISPR Therapeutics, and the Broad Institute control most trials, and the AI algorithms used to predict gene targets and optimize patient safety in these trials, are calibrated for predominantly white populations in high-resource settings. Such trial design limits the tool's applicability in low-income health systems; moreover, the lack of trial sites in SSA, as shown in Fig. 7, raises critical concerns. Without local trials, African communities are excluded from early therapeutic access and from shaping definitions of safety, efficacy, and acceptability that govern AI-CRISPR deployment [208]. Furthermore, the absence of clinical trials in SSA is not merely a logistical issue but also reflects challenges to epistemic sovereignty and persistent postcolonial power dynamics, evidenced by regulatory risks, weak infrastructure, and funding disparities [101].

Clinical Trial Sites in SCD*

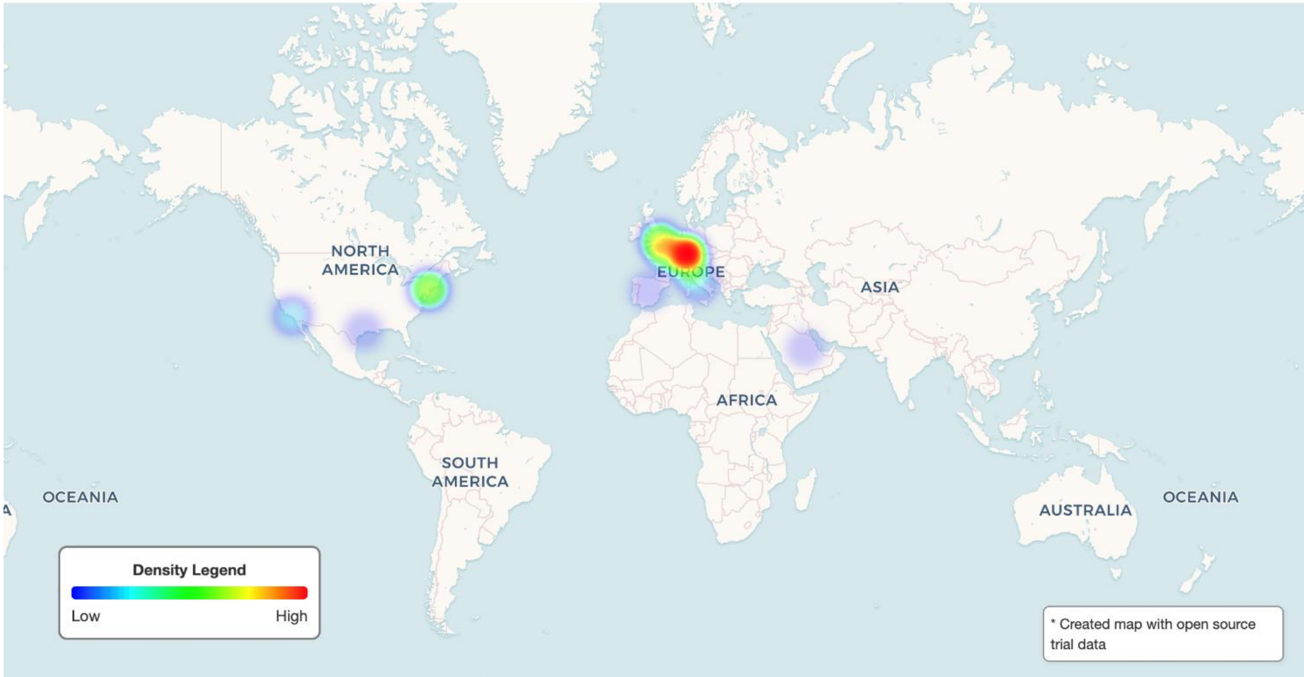

Fig. 6: SCD clinical trial sites. [28]

5.2.3 Manufacturing and IP: Patents, Markets, and Monopolies

With over 300 CRISPR therapies — many now integrating AI — progressing toward clinical maturity, examining SCD's current market access offers a valuable indicator of how future AI-CRISPR treatments might reach, or fail to reach, populations in need [215]. Currently, the production and distribution of AI-CRISPR SCD therapies are tightly controlled by a small number of Northern firms. Vertex, CRISPR Therapeutics, and Editas dominate patents related to delivery vectors, CRISPR enzymes, and proprietary AI toolchains [216]-[219]. Of more than 11,000 CRISPR-related patent families filed globally, less than 1% originate from African entities [218], [219]. Fig. 7 illustrates the global distribution of CRISPR patents, including SCD patents, which are heavily concentrated in the US and China [89], [206].

This concentration of IP rights and manufacturing capabilities restricts SSA's ability to locally produce or adapt CRISPR therapies, deepening reliance on external firms.[29] Moreover, SSA's ability to locally produce or adapt CRISPR therapies is further constrained by funding asymmetries that concentrate financial resources in Northern institutions, leaving African researchers and manufacturers with limited means to build autonomous

[28] Map data sources [92], [209]-[214].

[29] IP refers to intellectual property, protected by patents [220].

innovation pipelines. Even global health equity initiatives, such as the NIH-Gates Foundation's US$200 million gene-editing program, largely fund established labs in the Global North [221]-[224]. Such dependency limits SSA's capacity to exercise epistemic sovereignty over therapeutic innovation, as decisions about pricing, access, and adaptation are made elsewhere. With per-patient costs exceeding US $2 million, AI-CRISPR therapies remain far beyond the reach of most SSA health systems [225]. This funding architecture reflects deeper postcolonial hierarchies that have shaped global health systems, concentrating financial power in the North, and structuring CRISPR innovation in ways that sustain SSA's dependency. While CRISPR's cost is a barrier, prices will likely decline as the technology matures, making it a priority to advance AI-CRISPR's foundational development to enable future access.

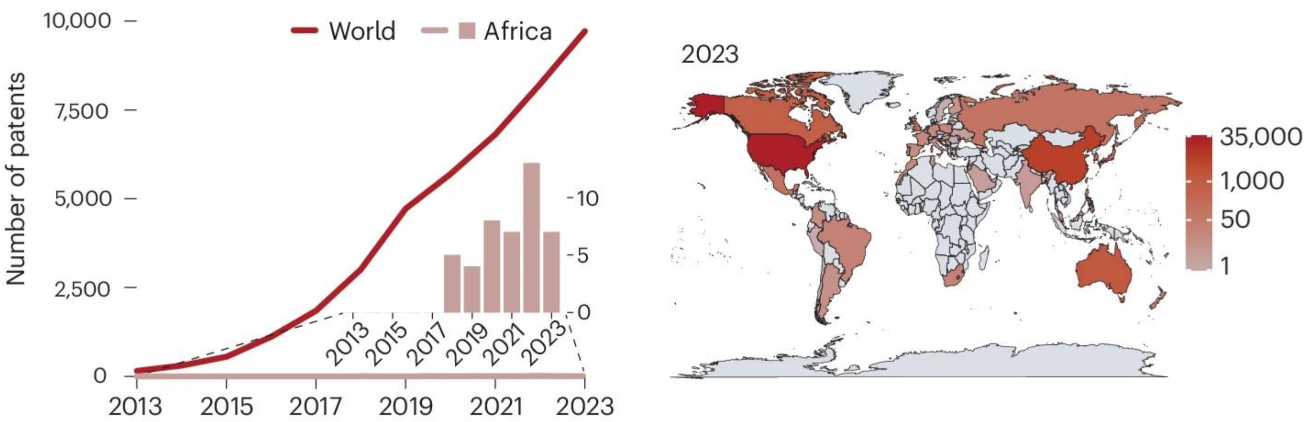

Fig. 7: Distribution of CRISPR patents by 2023 [145].

5.3 Infrastructure Dependencies: How AI-CRISPR Deepens Structural Reliance

Western dominance over funding networks, clinical research, market distribution, and proprietary patents already limits SSA's access to and agency over AI-CRISPR therapies. Yet these challenges are compounded by entrenched infrastructural disparities that further restrict the region's capacity to adapt and deploy gene editing technologies. Gene editing is not only a biomedical process but an infrastructure-intensive endeavor, requiring sophisticated physical, digital, and regulatory systems to function effectively [226]. These include high-throughput genomic sequencers, computational modelling environments, specialized wet labs, and cold-chain distribution networks, all of which are limited or unevenly distributed across SSA [93], [227].

AI-CRISPR systems amplify these demands. Their integration requires even more advanced infrastructure: GPU-powered computational clusters, cloud-based storage platforms, secure genomic databases, and highly trained personnel to maintain and interpret complex bioinformatic pipelines [228].[30] Such resources remain disproportionately concentrated in the Global North, reinforcing structural dependency and limiting the epistemic and operational autonomy of African research institutions. These infrastructural disparities are not merely technical or economic gaps; they reflect structural conditions that have long concentrated scientific capacity and technological authority outside SSA, now limiting regional epistemic sovereignty over genomic innovation.

5.3.1 Physical Infrastructure

Gene editing requires highly developed physical infrastructure including GMP-grade clean rooms, biomanufacturing pipelines, high-throughput sequencing machines [230]. Establishing a gene-editing facility can cost over US $100 million, while sequencing equipment like Illumina NovaSeq, ranges from $350,000 to $1 million per unit [231]-[233]. Across SSA, only a handful of centers have the capacity to support full ex vivo CRISPR workflows as highly controlled lab environments and trained personnel are a prerequisite to complete the technical editing process [93], [101]. Due to these prohibitive costs, SSA remains dependent on foreign institutions for genomic data storage and analysis as well, limiting regional agency over research and innovation [234]. Moreover, sustaining such infrastructure requires a skilled biomedical workforce, yet gaps persist due to scarce local training programs, outdated equipment, and talent migration to institutions in the Global North [235], [236]. While initiatives like H3ABioNet and regional genomics consortia have made important advances, they remain insufficient to support autonomous, locally controlled gene-editing ecosystems [237].[31] This dependency further constrains SSA's epistemic sovereignty over the design and application of AI-CRISPR technologies and reflect long-standing structural conditions that have concentrated scientific and technological capacity outside SSA, limiting regional agency [238].

[30] Bioinformatics is the use of computational tools to analyze biological data [229].
[31] H3ABioNet is a pan-African network that focuses on building African capacity for genomic research [237].

5.3.2 Computational Gaps: AI Infrastructure

AI-CRISPR systems add layers of computational complexity. State-of-the-art models require massive computing resources. Meta's LLaMA-3 70B parameter model, for instance, consumed over 6.4 million H100 GPU-hours during training, an effort estimated to cost tens of millions of dollars [167], [239]. [32] Even smaller-scale bioinformatics models can cost millions in cloud compute fees as they may involve training hundreds of millions of parameters [166], [241].[33] AI models rely on energy-intensive servers and cloud-based systems, infrastructure overwhelmingly controlled by providers in the Global North [93], [242]. Although platforms like Amazon and Google offer global access in principle, this access comes at a financial and governance cost that fosters dependency rather than true inclusion [93], [242], [243]. Such reliance limits SSA institutions' capacity to conduct independent AI-CRISPR research, shape locally relevant solutions, and exercise data sovereignty.

Even when African researchers contribute to international AI–CRISPR projects, their work often depends on external computational infrastructure for model validation, genomic comparison, and data visualization [244]. This dependence reinforces a structural exclusion from the computational core of knowledge production, where critical decisions about algorithms, infrastructure, and embedded values are made. In the AI–CRISPR era, achieving epistemic sovereignty in SCD interventions is not only about securing trial participation or equitable access to patents; it is also about who designs, controls, and operationalizes the computational systems that increasingly define how SCD therapeutic futures are imagined, validated, and deployed [104].

5.3.3 Epistemic Infrastructure?

Control over physical and computational infrastructure translates into control over epistemic authority, in that it shapes who count as credible actors, whose data is actionable, and whose interpretations carry authority in the biomedical domain [206]. Without the ability to generate, store, and interpret data locally, SSA scientists and institutions cannot independently define research priorities, validate models, or set safety standards [145]. Instead, they must work within paradigms and pipelines designed elsewhere, reinforcing dependency at every stage of AI-CRISPR deployment. This reflects a larger pattern of infrastructural dispossession in global science, where capital-intensive technologies reinforce hierarchies of epistemic power.

[32] A Graphics Processing Unit (GPU) performs thousands of parallel computations to accelerate AI machine learning tasks [240].

5.4 Ethical Governance: When Proceduralism Masks Epistemic Exclusion

The Global North's dominance over AI-CRISPR research and infrastructure not only sidelines African agency and epistemologies but also extends to the design and governance of these therapies. CRISPR is governed by a complex array of frameworks, including national agencies, international ethical guidelines from regulatory bodies, and institutional oversight mechanisms, all designed to ensure its safe and responsible development [245] . Global bioethics frameworks for gene editing, shaped by the WHO, UNESCO, FDA, privilege procedural values: autonomy, transparency, informed consent [51], [246]. These standards reflect liberal individualist moral philosophies that may not align with SSA contexts, where health is often understood relationally, and risks are intergenerationally negotiated [134], [135], [245], [247]. It's important to note that current CRISPR regulations focus on gene-editing mechanisms themselves, so for now, the nascent integration of AI is governed through these existing gene editing standards that shape tool design, deployment, and evaluation.

When Northern procedural ethics for gene editing applications, like SCD therapies, are applied without adaptation, they can obscure local moral vocabularies and reduce complex community values to formal compliance checklists [97], [131], [208]. By privileging procedural ethics rooted in Western liberal principles, these frameworks reproduce epistemic exclusion and overlook relational ethics, eroding trust, participation, and awareness of intergenerational risks [210]. This disjuncture not only reflects a cultural mismatch but actively reproduces epistemic exclusion by sidelining relational ethics — those grounded in trust, participation, and intergenerational responsibility [209]. As a result, what is lost is not merely alignment with local values, but the conditions necessary for equitable and sustainable innovation. For AI-CRISPR interventions in SSA, this means that ethical governance tends to be reactive and externally imposed, rather than collaboratively co-designed with the communities most affected [249], [250].

These patterns of exclusion extend beyond ethical frameworks to the very systems that govern the technical deployment of AI–CRISPR therapies. Regulatory asymmetries compound these challenges [68], [251]. Because most SSA regulatory agencies lack the capacity to assess CRISPR therapies independently as they lack reference labs for vector batch-release, off-target analytics, or AI model validation, they defer to Northern authorities [68], [101], [251]. This form of governance, lacking material capacity and epistemic inclusion, leaves SSA with little meaningful voice in setting the terms of AI-CRISPR deployment [252].

5.5 Algorithmic Accountability: Data, Design, and the Politics of AI-CRISPR

Lastly, a fundamental issue in the use of AI-CRISPR for SCD lies in who controls the genomic knowledge that informs therapeutic design. By integrating AI systems reflective of hierarchical data and design assumptions, the use of AI in gene editing reconfigures the terms under which knowledge is produced and authorized. As a result, algorithmic design becomes a site of epistemic contestation because of model numerical stability, bias, and truthfulness.

5.5.1 Data Biases and Genomic Underrepresentation

Generally, the reliability of AL/ML model outputs hinges on the quality and representativeness of input data. As Menard succinctly puts it, "for garbage in, you get garbage out" [253].[34] In the case of genomic applications, biased or incomplete data can produce faulty outputs with serious clinical implications [253]. Even when designed with ethical intentions, AI systems frequently reflect and reinforce underlying historical or societal biases due to imbalanced training data, leading to less accurate predictions of outputs [255], [256]. This is particularly concerning healthcare, where disparities in training data often translate into unequal diagnostic or therapeutic performance [256].

As many diseases are genetically heterogeneous and vary significantly across populations, diverse, representative datasets are essential in genomic research [253]. Yet most global genomic repositories, including databases used for CRISPR SCD studies, overwhelmingly skew toward individuals of European ancestry [110], [257]. As of 2020, over 90% of genome-wide association studies (GWAS) had been conducted on populations of European descent, while African genomes comprised less than 2% of public genomic databases [172], [258], [259]. This incongruity can result in algorithmic underestimation, erroneous pathogenic conclusions, and most importantly, lead to misinformed interventions. Studies show AI models trained solely on "Eurocentric" data may entirely overlook critical genomic features in African and Indigenous populations [257], [110], [260]. If AI-CRISPR systems cannot reliably predict outcomes for African genomes, their clinical deployment risks reduced efficacy, increased harm, and epistemic exclusion in which SSA SCD patients' biological realities are systematically mischaracterized by dominant scientific infrastructures.

[34] Garbage refers to biased data [254].

5.5.2 Opacity and Design: Embedded Algorithmic Value:

AI model architectures themselves are not biased, but the assumptions encoded in their design, such as loss functions, thresholds, target outcomes, reflect institutional and cultural priorities [261].[35] In CRISPR applications, these choices shape what constitutes "acceptable risk" or "therapeutic success" [253], [262], [263]. Most guide RNA design engines, designed by large biotech companies in the North, are proprietary, preventing external audit or local recalibration by communities most affected by these standards, that being SSA SCD patients in this case of AI-CRISR intervention [81].

These assumptions encode value judgments about what constitutes safety, what outcomes are prioritized, and how biomedical success is defined, shaped by specific institutional and cultural values, often centered in Northern institutions with limited sensitivity to local contexts [204], [212]. Overall, this opacity transfers decision-making power from local clinicians and communities to distant servers and regulatory boards, embedding external priorities in interventions intended for SSA patients. The result is technocratic health governance that reduces political and cultural concerns to algorithmic metrics, obscuring the ethical stakes of deploying AI–CRISPR for diseases like SCD [128], [264].

5.6 Achieving Epistemic Sovereignty in Gene Editing?

The SCD case shows that AI-CRISPR integration does not simply introduce technical improvements: it reconfigures existing asymmetries into new algorithmic, infrastructural, and legal dependencies. To democratize AI-CRISPR and support health equity, genomic innovation must be acknowledged as a matter of technological diffusion, but of epistemic redistribution [111]. This includes investing in regional genomic and AI computing hubs, enabling algorithmic co-design, and creating governance frameworks attuned to local values and capacities with affected communities and stakeholders [24], [146], [265]. Only then can AI-CRISPR move from frontier science to ethically legitimate, globally equitable health intervention [266]. The following section develops this argument further by synthesizing these empirical insights into a normative critique of how AI-CRISPR is governed — and how it might be governed in the service of epistemic sovereignty [267].

[35] These technical assumptions are key design choices that shape how a system defines success and its actions [261].

Section 6: Discussion

Building on the preceding case study of AI-CRISPR interventions for SCD in SSA, this section synthesizes key empirical insights into a critical normative reflection. While Section 5 demonstrated how current trajectories in genomic development reinforce epistemic exclusion, this section explores the structural conditions that sustain these inequities and identifies pathways toward equitable AI-CRISPR governance. Framed through the lens of epistemic sovereignty, this section analyzes how AI-CRISPR entrenches or challenges historical asymmetries, reshapes knowledge production and governance, and influences prospects for health equity in SSA. It addresses the dissertation's research questions and outlines pathways toward more inclusive genomic futures, given AI's role as an epistemic arbiter.

6.1 Reframing the Problem

The deployment of AI-CRISPR for SCD in SSA largely reproduces historical asymmetries in genomic data, infrastructure, and ethical authority [100]. Despite SSA's high burden of SCD, regional researchers and communities are excluded from defining treatment priorities, designing studies, or operationalizing results [268]. Critical gene editing infrastructure remains limited, while funding, IP rights, and technological control are concentrated in the Global North [141], [270]. CRISPR's political economy remains tied to Northern patents, venture capital, and industry strategies, often diverging from SSA's public health needs [59], [270].

AI systems integrated into CRISPR pipelines compound these inequities. Proprietary architectures, Eurocentric training data, and market-driven incentives prioritize Northern populations and health priorities, excluding African genomic realities from both datasets and algorithmic assumptions [125]. This exclusion is twofold: underrepresentation in data and lack of infrastructural autonomy. Even open-access collaborations or multilateral initiatives, while promising, can reinscribe hierarchies without mechanisms for co-authorship, benefit-sharing, or regional governance [271], [272].

AI-CRISPR for SCD reflects systemic exclusion and a failure of epistemic sovereignty, denying regional agency over genomic knowledge, priorities, and futures. This exclusion is not incidental. SSA's limited agency in biotechnology stems from chronic underinvestment, externally imposed IP regimes, and weak translational ecosystems — rooted in colonial legacies and enduring global health inequities. [248], [273]. Epistemic sovereignty highlights that these exclusions are structural, sustained by historical and ongoing hierarchies that define whose knowledge counts and who holds authority over biotechnological futures [276]. These dynamics highlight that AI is not solely a technical tool but an epistemic force shaping whose knowledge matters and which health futures are prioritized, a theme explored in the following subsection [277], [278].

6.2 The Role of AI in Gene Editing: From Tool to Epistemic Arbiter

AI's integration into CRISPR pipelines transforms gene editing from a purely technical process into an algorithmically mediated system of decision-making, one that shapes whose health needs count, which risks are prioritized, and how precision is defined. AI systems act as gatekeepers in gene editing, scoring genetic changes, assessing off-target risks, and shaping the norms of what counts as precision, safety, and validity [279]. In doing so, these systems implicitly decide which outcomes matter and which populations are made visible or invisible in genomic innovation [280], [281]. AI-CRISPR architectures embed specific values and commercial incentives attuned to Northern populations and markets, deepening inequities by hardwiring underrepresentation into both data and algorithmic assumptions in regions like SSA [282], [283]. In doing so, AI becomes a key force in perpetuating epistemic exclusion in global gene editing, consolidating control over SSA's genomic future in the hands of external actors.

While initiatives such as H3Africa, the African Medicines Agency, and regional pilot programs show potential for reshaping governance toward greater inclusivity, their impact remains constrained without structural reforms that redistribute authority over AI standards, data governance, and algorithmic design. AI-CRISPR's role in global health innovation therefore reflects a broader struggle over who sets scientific priorities and defines ethical legitimacy in genomic futures. Without structural shifts in authority over AI standards, data governance, and algorithmic design, efforts to contest these inequities will fall short of true epistemic sovereignty in genomic innovation.

6.3 Reorientating the Politics of AI-CRISPR

AI-CRISPR, as currently developed, entrenches exclusionary epistemic hierarchies rather than advancing SSA's epistemic sovereignty in genomic innovation. Northern institutions continue to define the standards, models, and regulatory frameworks that govern AI-CRISPR, leaving SSA with little authority over the design, data governance, or ethical norms shaping these technologies.

Achieving epistemic sovereignty demands more than token inclusion in datasets or collaborations; it requires redistributing authority across the entire AI-CRISPR pipeline through co-authorship, regional validation, infrastructural autonomy, and governance grounded in local contexts. The goal is not simply to make AI-CRISPR "more ethical," but to decolonize how genomic knowledge is produced, validated, and applied. Realizing this vision requires governance frameworks that enable SSA institutions not only to shape AI-CRISPR standards and deployment, but to lead in designing, regulating, and overseeing these technologies on their own terms.

SSA's emerging genomic initiatives — to be discussed in the next subsection — show potential for progress, yet they remain constrained without systemic shifts in who controls genomic knowledge and technology. The potential of AI-CRISPR to reconfigure epistemic sovereignty ultimately hinges on dismantling these structural asymmetries and enabling affected communities to exercise genuine authority over their genomic futures.

6.4 Steps Towards Epistemic Sovereignty

Achieving equitable AI-CRISPR futures requires more than technical innovation; it demands structural change. Longstanding patterns of unequal investment, data monopolization, and externally controlled IP continue to limit SSA's agency in genomic innovation [199], [284]. Falling costs and growing precision in gene editing present new opportunities to treat diseases endemic to SSA, yet these benefits will remain unevenly distributed unless regional research co-authorship, infrastructural autonomy, and participatory governance are prioritized [285], [244].

Emerging SSA scholarship shows initiatives like H3Africa, the African Medicines Agency (AMA), and regional bioethics platforms have laid foundations for more context-sensitive, accountable genomic governance [202], [207], [265], [286]. For example, Nigerian scientists are exploring CRISPR hemoglobin treatments, while the African Health Organization recently announced a regional exa-cel pilot for SCD patients [265], [287], [192]. Similarly, Toto et al.'s reflections on the Malawi Cryptosporidium study emphasize that successful biomedical programs in SSA often depend on careful planning, adaptive oversight, and sustained responsiveness to local contexts [288].

Concurrently, calls to decolonize global health research are gaining traction, as scholars challenge the conditions and hierarchies that exclude Global South researchers from defining "legitimate science" [274], [275], [289]. For instance, the genomic sovereignty movement, first articulated in Latin America and now echoed in SSA, demands greater control over "Indigenous and minority genetic data," — how it is produced, interpreted, and applied [39]. As policymakers and geneticists increasingly advocate culturally sensitive, community-driven research, epistemic sovereignty emerges not as an abstract ideal, but as a growing practice to reclaim authority over data use, regulation, and research agendas [250], [251], [292], [293].

Although these efforts highlight SSA's growing potential for regional leadership, genomics remains shaped by enduring postcolonial dependencies. Epistemic sovereignty is not a fixed endpoint but a dynamic struggle to reclaim authority over knowledge, technology, and governance. Realizing this vision requires transforming institutional and infrastructural foundations through participatory, transparent governance models, including local ethics boards, regional data trusts, and community-led oversight grounded in context, historical awareness, and accountable algorithmic design [294]. Co-creating research agendas with underserved SSA populations, ensuring benefits are returned through accessible treatments, and fostering interdisciplinary collaboration will aid this effort [298]. Targeted interventions through public-private partnerships, localized policy, and regional investment are essential to ensure AI-CRISPR development in SSA disrupts, rather than perpetuates, colonial patterns of healthcare [295].

6.5 Operational Pathways

Rather than propose sweeping policy directives for genomic governance in SSA, this dissertation explores how AI-CRISPR reshapes scientific knowledge, power, and representation. In materializing epistemic sovereignty as a tangible asset, it also offers forward-looking recommendations that support regionally driven, context-sensitive pathways for equitable AI-CRISPR governance. By consolidating insights from genomic scholars including Ogaugwu, Happi, Kiyaga, and Abkallo, the goal is to inspire future work that builds on regionally grounded SSA scholarship, avoiding the imposition of external solutions and ensuring interventions emerge from local expertise and priorities. While this dissertation focuses on AI-CRISPR in SSA, strategies to enhance biotechnological accessibility must reflect the region's internal diversity. SSA is not a monolith; including scholars from across the region helps ensure that efforts to decolonize genomic therapies engage its varied epistemologies, regulatory contexts, and health priorities. Accordingly, Fig. 8 synthesizes key gene therapy recommendations tailored for SSA, while Fig. 9 maps genomic capacity-building priorities across LMICs based on Doxzen et al.'s work [101].

Recommendation Area	Key Actors / Examples	Strategic Focus	Indicative Metrics	Time Horizon	Safeguards/Notes
1. African Genomics Research Networks	H3Africa, African Society for Human Genetics	Promote African-led genomic research, capacity building, and ethical frameworks prioritizing local governance	Number of African-led projects; training workshops conducted	Short-medium (1-5 yrs)	Regional ethics committees oversight
2. Regional Bioethics & CRISPR Governance	AUDA-NEPAD (African Union Development Agency - New Partnership for Africa's Development), African Union policy bodies	Develop regional policy and regulatory structures for CRISPR; reduce dependence on Western governance models	Policies/regulations enacted; regional ethics boards established	Short (1-3 yrs)	Align with African Union/WHO ethics standards
3. Community-Based Genetic Research	AfriGen (South African biotech firm), local biobanking initiatives	Facilitate grassroots engagement and ethical deliberation grounded in local cultural and epistemic norms	Community consultations held; local biobanks supported	Short-medium (1-5 yrs)	Community consent protocols mandatory
4. Collaborative Research Models	African universities, Novartis-SSA partnership (pharmaceutical firm) , transnational biotech collaborations	Create equitable partnerships to ensure African priorities shape CRISPR research and innovation trajectories	Number of co-authored studies; funding allocation ratios	Medium (3-7 yrs)	Transparent MoUs (Memorandum of Understanding); public reporting of partnership terms
5. Infrastructure Redistribution	H3ABioNet expansion, South-South partnerships	Develop regional genomic and AI computing hubs; advance sustainable financing to move beyond donor dependency	Genomic/AI hubs established; % funding from non-donor sources	Medium-long (3-7+ yrs)	Shared regional governance; anti-monopoly safeguards
6. Algorithmic Co-Design	SSA AI researchers, open-source initiatives	Require SSA co-authorship and co-ownership in RNA libraries, risk models, AI tools; enable open-source auditing for local fit	% of AI tools co-designed; open audit frameworks implemented	Medium (3-7 yrs)	Open-source licensing; data sovereignty protections
7. Plural Ethics	Community groups, regional ethics boards	Embed collective care, intergenerational risk awareness, and community consent into trial protocols and regulatory design	Trials with plural ethics model; protocols with intergenerational input	Short-medium (1-5 yrs)	Mandatory ethical audits; culturally appropriate consent forms
8. Equitable Access and Delivery	Global Fund, GAVI (Vaccine Alliance), WHO partnerships	Ensure gene therapies are accessible and affordable in LMICs through pooled procurement, tiered pricing, and local delivery	Therapies delivered via pooled schemes; cost reductions achieved	Long (5-10 yrs)	Pricing transparency; access monitoring

Fig. 8: Key SSA recommendations, consolidated from SSA scholarship [145], [202], [296], [297]

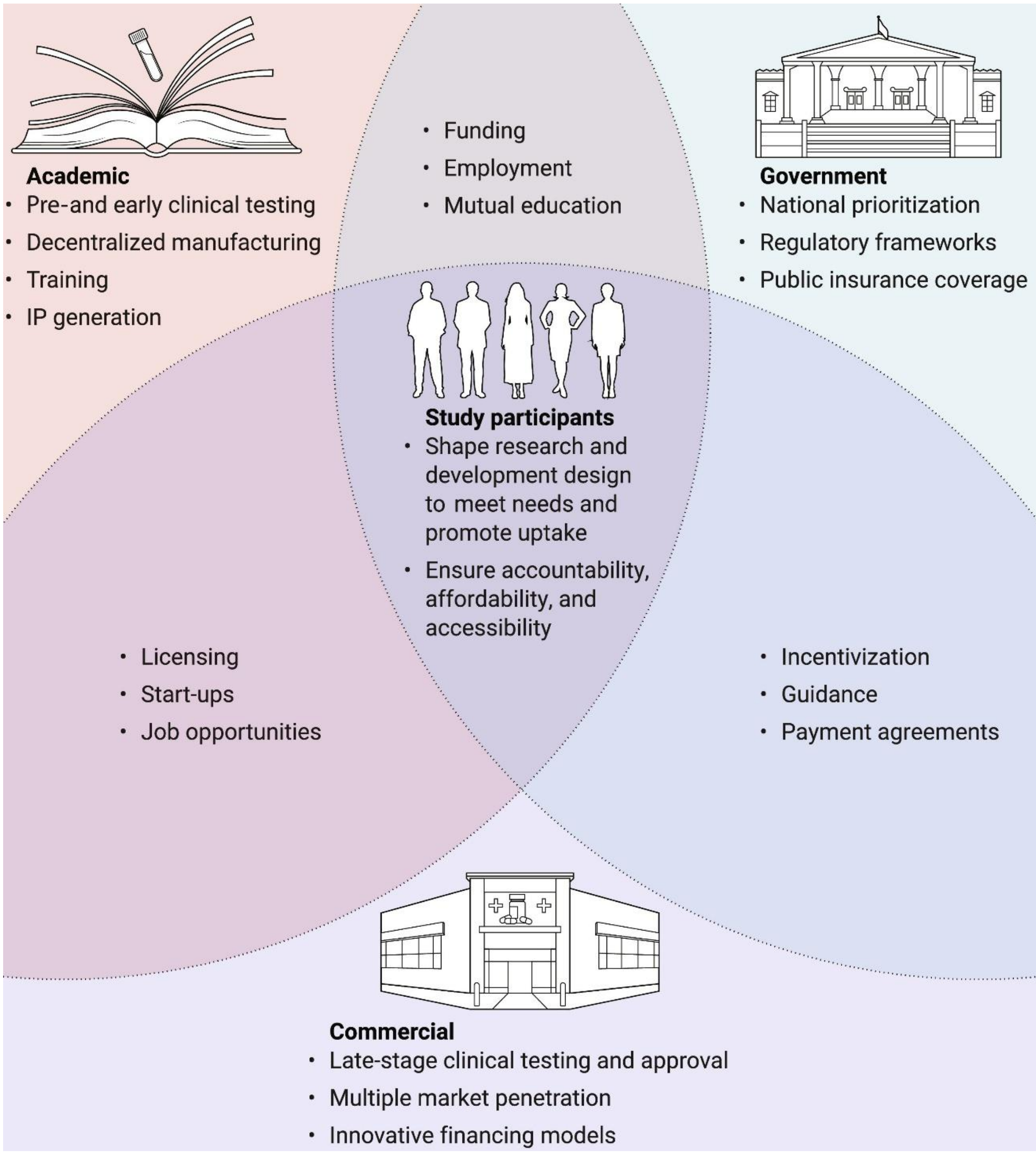

Fig. 9: Spheres for building LMIC's gene editing capacity [101]

Building on these figures, Fig. 10 translates these high-level recommendations into actionable steps already piloted or implemented within African health systems. It offers concrete, context-sensitive measures that operationalize the priorities outlined in figures 8 and 9, providing a practical roadmap for advancing equitable AI-CRISPR governance and a basis for future research. However, these operational pathways are not exhaustive and represent initial steps that must be expanded through regionally led research, iterative governance processes, and sustained collaboration.

#	Priority action: transfering existing measures to AI-CRISPR	Concrete precedents (proof-of-concept)	Why it advances epistemic sovereignty
1	**Continental Gene-Editing board inside the African Medicines Agency (AMA)** – issues one binding decision on trials & post-market safety for all African member states.	AMA Governing Board began constituting advanced-therapy committees in 2024 **[298]**; WA-MRH's joint dossier reviews show the workflow template **[299]**	Provides a legal pipeline to facilitate external sovereignty by requiring regional sign-off before trials or marketing authorisation. Let African regulators set the gold standard and affirm their position towards Western countries
2	**Couple national CRISPR rules with community-embedded ethics boards.** Every country that adopts gene-editing guidelines must require a local cummunity prevalance.	Nigeria, Kenya & Malawi genome-editing guidelines **[300]**; Malawi's COMREC model blends scientists and village reps	Marries clear legal and local authority to basic organisation oversight limiting elite or foreign dominance.
3	**African genomic data trusts committee and recursive algorithmic audits.** Raw genomes stay on African servers; any AI tool must publish error rates on African ancestry data.	H3Africa Data & Biospecimen Access Committee (DBAC); African Data Ethics charter calls for algorithmic audits **[301]**	Locks in local data control *and* forces transparency from proprietary AI scoring engines. Also allows sound collaboration and comparison with current AI-CRIPSR models.
4	**Tech-transfer hubs and continent-wide IP flexibilities.** Extend the WHO/MPP mRNA hub model to Cas9, delivery vectors & AI models, backed by AMA/AMRH TRIPS policies.	WHO–MPP mRNA hub at Afrigen, SA, now training 14 LMIC "spokes"; policy papers urge AMA to deploy TRIPS flexibilities for essential medicines **[302]**	Couples hands-on know-how with legal tools that guarantee freedom-to-operate for regional priorities such as SCD therapies.
5	**Enhancing funding in domestic and regional opportunities for sequencing labs and AI compute.** Leverage a global Genomics & AI Sovereign Fund.	Africa CDC's US $100 m Pathogen Genomics Initiative building 20+ sequencing hubs; Illumina's $20 m pledge and other matching grants illustrate blended financing **[303]**	Provides stable, home-grown capital so infrastructure and talent pipelines outlive donor cycles.

Fig. 10: Actions for inclusive AI-CRISPR futures

Together, these figures highlight the need to embed gene editing within frameworks of epistemic sovereignty, ensuring interventions are inclusive, locally led, and technically sound. This requires deeper ethical reflection on AI's role in genomics, as AI is not a neutral tool but an epistemic arbiter shaping which biological futures are pursued. Epistemic sovereignty must evolve with just algorithmic governance, enabling communities to access, interrogate, and co-produce these technologies. Specifically, AI-CRISPR governance must prioritize transparency, contextual relevance, and participatory accountability, with models judged not by technical benchmarks alone, but by their impact on marginalized populations through continuous algorithmic audits with meaningful SSA oversight [304], [305]. This vision aligns with scholars like Birhane, D'Ignazio, and Klein, who argue ethical AI must be grounded in relational accountability and knowledge co-production, not extractive data practices or post-hoc audits that leave systemic inequities intact [167], [306].

While epistemic sovereignty exposes structural asymmetries, addressing challenges posed by proprietary technologies, closed algorithms, and transnational regulatory regimes in this field demands complementary legal, institutional, and infrastructural reforms.

Advancing equitable genomic futures demands sustained collaboration among SSA bioethicists, AI developers, and community stakeholders to co-develop governance frameworks that democratize control over data and AI infrastructure alike. Ensuring AI-CRISPR fosters inclusive innovation for health equity means advancing its technical safety and epistemic sovereignty, in tandem, as genomics and AI coevolve [301]. Only then can AI-CRISPR become a vehicle for inclusive genomic innovation in the region.

Section 7: Conclusion

This dissertation has examined the convergence of AI and gene editing through the lens of epistemic sovereignty, using the case of CRISPR interventions for SCD in SSA as a critical site of inquiry. Although AI-CRISPR is celebrated as a genomic breakthrough, this analysis shows its development and deployment remain deeply entangled in global asymmetries of knowledge, infrastructure, and governance. These asymmetries reflect longstanding patterns of scientific authority shaped by colonial legacies and sustained through proprietary systems, exclusionary regulatory regimes, and algorithmic opacity, now reconfigured through AI's role in genomics.

Drawing on postcolonial STS, this dissertation has argued that AI is not a neutral amplifier of CRISPR's technical capabilities, but an epistemic force that encodes assumptions about what counts as valid knowledge, which populations are prioritized, and who exercises authority over biomedical futures. AI-CRISPR research, clinical trials, infrastructure, distribution networks, and regulatory frameworks for SCD are largely centered in the Global North, with their algorithms developed by Western biotech firms and trained on Eurocentric, predominantly white datasets, reinforcing structural biases in genomic innovation. In this context, SSA remains disproportionately excluded, not only from the material benefits of genomic innovation, but from shaping its purpose, ethics, and direction.

This study acknowledges several limitations: it lacks primary ethnographic interviews or country-specific analyses of biotechnology landscapes in SSA, which would likely enrich the analysis. While epistemic sovereignty provides a valuable lens for revealing the asymmetries embedded in the AI-CRISPR lifecycle, its practical limitations, including challenges in global interoperability, suggest that complementary frameworks could enhance future applicability across diverse transnational contexts. Lastly, some may reasonably argue that SSA should avoid dedicating limited resources to "speculative" high-tech genomic therapies [307]. While investing in foundational health infrastructure is essential, engaging with emerging genomic technologies is not mutually exclusive and can help ensure that SSA shapes, rather than inherits, the future of global health innovation. As such, this dissertation does not ask whether AI-CRISPR should be pursued, but *how* — on whose terms, with whose oversight, and for whose priorities. The challenge lies not in technology itself but in the governance structures shaping its development and use. As AI-CRISPR's technical refinement is inevitable, the focus must be on the underlying forces guiding AI-CRISPR's application in SSA.

AI-CRISPR gene editing represents a promising horizon for scientific innovation, with the potential to cure millions of painful genetic diseases and thereby improve global health. However, in practice, AI-genetic technologies like CRISPR remain inextricably embedded in existing scientific systems of exclusivity. Without reconfiguring epistemic authority across the innovation pipeline through collaborative, pluralistic, and accountable governance models, AI-CRISPR risks reinforcing patterns of epistemic exclusion already present in biomedicine. Nonetheless, the future of AI-CRISPR —and AI-driven gene editing more broadly — is still unfolding. Whether these tools foster inclusive innovation and advance health equity or deepen epistemic asymmetries, will depend on who governs their development and whose priorities guide their global use. For SSA and other historically marginalized regions, the challenge is not only accessing frontier biotechnologies but reclaiming the authority to define their development, application, and governance. By centering epistemic sovereignty as both a guiding principle and an operational necessity, emerging biotechnologies can promote both equitable and accountable global health futures.

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